Phase-kernel classification, causal-diagonal one-use, and exact off-diagonal reduction

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

first issued 2026-07-24 · this version issued 2026-07-24 · v1.0

1 Purpose and scope

R-071 closes the complete fixed-floor linear production frame and proves that the nonlinear terminal current square has a genuine kernel leakage. This note identifies that kernel exactly, proves the strict-past one-use estimate for the matched same-shell part of the leakage, and expands the full terminal leakage into that diagonal plus one explicit off-diagonal remainder.

The diagonal estimate is cutoff-uniform and uses one Cameron–Martin allocation, one terminal sextic allocation, and one integrable random constant. It does not close the full nonlinear subgate: the off-diagonal remainder is nonzero and can dominate the diagonal. The next theorem must triangularly group that remainder together with the range-visible terminal square. No finite-energy extension, controlled-shell one-use theorem, or Nelson estimate is asserted here.

All statements use the hash-pinned A1 production coefficients, the R-050/A7 common real-even scalar regulator, the R-066 strict sharp-cube filtration, a fixed positive density floor, and regular whole-shell predictable controls. The finite-low boundary is retained.

2 The stacked production frame has an exact gauge kernel

Write a field value as $x = (u, s) \in \mathbb{C}^2 \oplus \mathbb{C}$. For the three embedded Pauli generators put

$$p_r(x) = 2(\sigma_r u, 0), \qquad v_r(x) = p_r(x) - 2q_r(x)x, \qquad M_r(x) = [p_r(x), v_r(x)], \tag{2.1}$$

and define the stacked current map

$$T_x b = (M_r(x)^T b)_{r=1}^3. \tag{2.2}$$

The production matrix Q_{II} is positive definite, so weighting (2.2) does not change its kernel.

Let $J_D(u, s) = (iu, 0)$ and $J_S(u, s) = (0, is)$. Then

$$\boxed{\ker T_x = \begin{cases} \mathrm{span}_{\mathbb{R}}\{J_D x, J_S x\}, & u \neq 0,\ s \neq 0, \\ \mathrm{span}_{\mathbb{R}}\{J_D x\} \oplus (\{0\} \times \mathbb{C}), & u \neq 0,\ s = 0, \\ \mathbb{R}^6, & u = 0. \end{cases}} \tag{2.3}$$

Indeed, the three vectors $\sigma_r u$ are mutually orthogonal in the real pairing and have common squared norm $|u|^2$. Their real orthogonal complement is therefore $\mathbb{R}, iu$. The Pauli/Hopf identity

$$\sum_{r=1}^{3} m_r \sigma_r u = |u|^2 u, \qquad m_r = u^\dagger \sigma_r u. \tag{2.4}$$

shows that the v_r conditions additionally impose $\mathrm{Re}(\bar{t}s) = 0$ on a singlet displacement t. This proves (2.3), including both degenerate strata. It also proves that a globally smooth inverse or fixed-rank pseudoinverse cannot be part of the argument.

The realified Pauli matrices commute with both J_D and J_S. Hence

$$M_r(x)^T J_\alpha x = 0, \qquad \alpha \in \{D, S\}. \qquad (2.5)$$

Differentiating (2.5) at z along a gives

$$DM_r(z)[a]^T J_\alpha z + M_r(z)^T J_\alpha a = 0. \qquad (2.6)$$

For

$$E_r = M_r(z+a) - M_r(z) - DM_r(z)[a]. \qquad (2.7)$$

one obtains the exact nonlinear gauge identity

$$\boxed{E_r^T J_\alpha (z+a) = -DM_r(z)[a]^T J_\alpha a.} \qquad (2.8)$$

The R-071 fixture is precisely this mechanism. Its recorded vector $n = (-1, 0, 0, 1, -1, 0)$ equals $J_D(z+a)$, not an accidental exceptional nullvector. Equation (2.8) reproduces the positive leakage slope

$$n \cdot \sum_r E_r Q_{\mathrm{II}} M_r(z)^T y = \frac{27(6e^2 + 22e + 27)}{400(e+3)^3} > 0. \qquad (2.9)$$

3 Inverse-free local completion

Let $T = T_x$, $H = T^* Q_{\mathrm{II}} T \succeq 0$, $d \in \mathbb{R}^6$ denote stacked terminal-current data in its natural target coordinates, and let $k \in \mathbb{R}^6$ be a derivative-displacement coefficient. For every $0 < \vartheta < 1$ and $\tau > 0$, completion with the strictly positive matrix $\vartheta H + \tau I$ gives

$$\begin{aligned}\frac{1}{2}|d + Tb|_Q^2 + k \cdot b \geq \frac{1-\vartheta}{2}|d + Tb|_Q^2 - \frac{\tau}{2}|b|^2 + \frac{\vartheta}{2}|d|_Q^2 \\ - \frac{1}{2}(\vartheta T^* Q d + k)^T (\vartheta H + \tau I)^{-1} (\vartheta T^* Q d + k). \qquad (3.1)\end{aligned}$$

No singular frame inverse occurs, and a fraction $1 - \vartheta$ of the terminal square remains. On an exact gauge direction, however, the last term can grow like $|k_{\mathrm{ker}}|^2/\tau$ as $\tau \downarrow 0$. Thus τ must be supplied by Cameron–Martin energy; (3.1) is an architecture, not by itself a closure.

For the actual frame set

$$w_r = M_r(z)^T y, \qquad k(z, a, y) = \sum_{r=1}^{3} E_r Q_{\mathrm{II}} w_r. \qquad (3.2)$$

The R-069 good/bad split gives the inverse-free global estimate

$$\boxed{|k(z, a, y)| \leq C_* |a|^2 |y|.} \qquad (3.3)$$

$$C_* = 24 \|Q_{\mathrm{II}}\|_{\mathrm{op}} (\sqrt{34} + 4\sqrt{2}) = 1.5397534378598672. \qquad (3.4)$$

For completeness, on $|a| \leq \theta s_e(z)$ Taylor's formula gives the safe constant $48\sqrt{8}\|Q\|/(1-\theta)$. On the complement, global frame Lipschitz continuity and $|z| < |a|/\theta$ give $24\sqrt{34}\|Q\|/\theta$. They agree at

$$\theta = \frac{\sqrt{34}}{\sqrt{34} + 4\sqrt{2}}. \qquad (3.5)$$

which proves (3.3)–(3.4) uniformly in the positive floor.

4 Strict-past diagonal one-use theorem

Fix a high block $j_0 \le j \le J$, $N_j = 2^j$, and write

$$a_j = K_j h_j, \qquad b_{j,i} = D_i a_j, \qquad z_{<j} = U_J + A_{<j}. \qquad (4.1)$$

Here h_j, a_j, and b_j are $\mathcal{F}_{j-1}$-measurable; U_J is the coherent terminal heat-value variable and is not spatially differentiated. Put

$$E_{r,j} = M_r(z_{<j} + a_j) - M_r(z_{<j}) - DM_r(z_{<j})[a_j]. \qquad (4.2)$$

and, with $G_{j,i} = D_i X_{<j}$,

$$\Lambda_j = \sum_{r,i} \int \langle M_r(z_{<j})^T G_{j,i}, Q_{\mathrm{II}} E_{r,j}^T b_{j,i} \rangle \, \mathrm{d}x. \qquad (4.3)$$

Use full-gradient constants

$$\|Da_j\|_2 \le C_\nabla N_j^{-1} \|h_j\|_{\mathcal{H}_j}, \qquad \operatorname{Tr} \operatorname{Cov} G_j(x) \le C_D N_j. \qquad (4.4)$$

The second trace is for the full eighteen-real-component vector: six target components times three spatial directions. If R-066's C_G is used one direction at a time, $C_\nabla = \sqrt{3} C_G$; the corresponding directional trace bounds are summed into C_D.

Equation (3.3), spatial Holder, and (4.4) give

$$|\Lambda_j| \le m_j x_j^{1/2} y_j^{1/3}. \qquad (4.5)$$

$$x_j = \|h_j\|_{\mathcal{H}_j}^2, \qquad y_j = \|a_j\|_6^6, \qquad m_j = C_* C_\nabla N_j^{-1} \|G_j\|_6. \qquad (4.6)$$

Apply sequence Holder once, with exponents $(2, 3, 6)$:

$$\sum_j |\Lambda_j| \le X^{1/2} Y_d^{1/3} R_\infty^{1/6}. \qquad (4.7)$$

$$X = \sum_j x_j, \qquad Y_d = \sum_j y_j, \qquad R_\infty = \sum_{j \ge j_0} m_j^6. \qquad (4.8)$$

One weighted AM–GM inequality then gives, pathwise,

$$\boxed{\sum_{j=j_0}^{J} |\Lambda_j| \le \eta X + \zeta_d Y_d + \frac{R_\infty}{432 \eta^3 \zeta_d^2}.} \qquad (4.9)$$

This is not shellwise Young with an arbitrary constant at every scale. It is one inequality with one accumulated random constant.

For a centered Gaussian vector g with covariance Σ,

$$\mathbb{E}|g|^6 = (\operatorname{Tr} \Sigma)^3 + 6 \operatorname{Tr} \Sigma \operatorname{Tr}(\Sigma^2) + 8 \operatorname{Tr}(\Sigma^3) \le 15 (\operatorname{Tr} \Sigma)^3. \qquad (4.10)$$

Therefore

$$\boxed{\mathbb{E} R_\infty \le 15 L^3 (C_* C_\nabla)^6 C_D^3 \frac{8}{7} N_{j_0}^{-3}.} \qquad (4.11)$$

Before conversion to the terminal sextic, the deterministic expected remainder in (4.9) is

$$\frac{5}{126} L^3 (C_* C_\nabla)^6 C_D^3 \eta^{-3} \zeta_d^{-2} N_{j_0}^{-3}. \qquad (4.12)$$

Only a sixth Gaussian moment is used.

4.1 Exactly one terminal sextic and one cutoff-independent constant

Pointwise,

$$Y_d \le \left\| \left(\sum_j |a_j|^2 \right)^{1/2} \right\|_6^6. \qquad (4.13)$$

The existing sharp-cube Littlewood–Paley theorem, with its actual constant C_{LP} rather than one, gives

$$Y_d \le C_{\mathrm{LP}}^6 \|A^{\mathrm{hi}}\|_6^6. \qquad (4.14)$$

For every $\alpha > 1$ there is $C_\alpha < \infty$ such that

$$|Z - U|^6 \le \alpha |Z|^6 + C_\alpha |U|^6. \qquad (4.15)$$

Choose $\zeta_d = \zeta/(\alpha C_{\mathrm{LP}}^6)$. If

$$U_* := \int \sup_J |U_J(x)|^6 \, \mathrm{d}x. \qquad (4.16)$$

then the vector martingale Doob inequality gives

$$\mathbb{E} U_* \le (6/5)^6 \mathbb{E} \|U_\infty\|_6^6 < \infty. \qquad (4.17)$$

Consequently (4.9) spends precisely $\zeta \|Z_J\|_6^6$ and leaves the single cutoff-independent integrable random constant

$$\boxed{C_{\eta,\zeta}(\omega) = \frac{\alpha^2 C_{\mathrm{LP}}^{12}}{432 \eta^3 \zeta^2} R_\infty + \frac{\zeta C_\alpha}{\alpha} U_*.} \qquad (4.18)$$

No Nelson exponential moment and no Malliavin norm is used.

4.2 Causalization is expectation-only

The terminal root in the algebra is $G = DX_J$, not G_j. At each spatial point condition on $\mathcal{F}_{j-1}$ enlarged by all future same-point value variables. Real-even covariance makes every fresh/future derivative increment centered and independent of that enlarged value sigma-field. Thus spatial Fubini gives exactly

$$\mathbb{E} \Lambda_j(G) = \mathbb{E} \Lambda_j(G_j). \qquad (4.19)$$

Equation (4.19) is not a pathwise replacement and not global field value–derivative independence. It fails for progressive within-shell feedback.

5 Exact off-diagonal remainder

The diagonal theorem is not the terminal theorem. To state the exact gap without dropping a finite-low boundary, start the high expansion at

$$z_0 = U_J + A_{<j_0}, \qquad A^{\mathrm{hi}} = \sum_{j=j_0}^{J} a_j, \qquad C^{\mathrm{hi}} = \sum_{\ell=j_0}^{J} b_\ell. \qquad (5.1)$$

The R-066 finite-low estimate retains the difference between U_J and z_0. For $z_{<j} = z_0 + \sum_{j_0 \le k < j} a_k$, set

$$E_j = M(z_{<j} + a_j) - M(z_{<j}) - DM(z_{<j})[a_j]. \qquad (5.2)$$

$$F_{k,j} = [DM(z_{\le k}) - DM(z_{<k})][a_j], \qquad k < j. \qquad (5.3)$$

Two telescopes give

$$\boxed{E_{\text{tot}} := M(z_0 + A^{\text{hi}}) - M(z_0) - DM(z_0)[A^{\text{hi}}] = \sum_j E_j + \sum_{k<j} F_{k,j}.} \tag{5.4}$$

Suppress generator and spatial-direction sums, put $w_0 = M(z_0)^T G$ and $w_{<j} = M(z_{<j})^T G$, and define

$$\begin{aligned}\mathcal{O} = &\sum_j \langle w_0 - w_{<j}, QE_j^T b_j\rangle \\ &+ \sum_j \sum_{\ell \neq j} \langle w_0, QE_j^T b_\ell\rangle \\ &+ \sum_{k<j} \sum_\ell \langle w_0, QF_{k,j}^T b_\ell\rangle. \end{aligned} \tag{5.5}$$

Then

$$\boxed{\langle w_0, QE_{\text{tot}}^T C^{\text{hi}}\rangle = \sum_j \langle w_{<j}, QE_j^T b_j\rangle + \mathcal{O}.} \tag{5.6}$$

The first sum is the matched diagonal covered by Section 4 after (4.19). The remainder $\mathcal{O}$ is exact and load-bearing. In the independent production fixture,

$$\text{LHS} = 0.0583409867702, \qquad \text{diagonal} = 1.42693028427 \times 10^{-5}, \qquad \mathcal{O} = 0.0583267174674. \tag{5.7}$$

so $|\mathcal{O}|/|\text{diagonal}| > 4087$. Calling it a small bookkeeping error is therefore false.

Terms in (5.5) with b_ℓ from a later shell can depend on intervening derivative innovations. Applying (4.19) to them before triangular expansion would require Malliavin derivatives of future controls. R-069 instead shows that accumulated past control currents and their mixed currents telescope when kept together with the terminal square. The missing compatibility lemma must prove, without double counting the R-071 linear payment, that (5.5) either returns to that telescope or satisfies a two-parameter largest-shell Carleson estimate with the range square retained.

6 Quantitative discriminator for the next root model

Let $X = \|A\|_{H^2}^2$ and $Y = \|A\|_6^6$. For $0 \leq \sigma \leq 1$, endpoint product estimates followed by Sobolev interpolation give

$$\boxed{\|A^2 DA\|_{H^\sigma} \leq C X^{(1+\sigma)/2} Y^{(2-\sigma)/6}.} \tag{6.1}$$

At $\sigma = 0$, use $\|A^2 DA\|_2 \leq \|A\|_6^2 \|DA\|_6$. At $\sigma = 1$, use $\|A\|_\infty^2 \lesssim \|A\|_{H^2} \|A\|_6$ and the Leibniz rule. Interpolation proves (6.1).

If a root model has norm M in $H^{-\sigma}$, termwise duality leaves room for one random constant exactly when $\sigma < 1/2$. Weighted Young then needs

$$M^{6/(1-2\sigma)}, \qquad \eta^{-3(1+\sigma)/(1-2\sigma)}, \qquad \zeta^{-(2-\sigma)/(1-2\sigma)}. \tag{6.2}$$

At $\sigma = 1/2$, (6.1) is the saturated product $X^{3/4} Y^{1/4}$ and leaves no exponent for an unbounded random norm. Therefore a raw $H^{-1/2-\delta}$ kernel-root model must gain more than δ derivatives from causal structure. If the gain is γ, its required moment is

$$\boxed{p = \frac{3}{\gamma - \delta}, \qquad \gamma > \delta.} \tag{6.3}$$

This is a discriminator for an absolute Sobolev fallback, not a no-go against signed range-square cancellation.

7 Why the local fixture is not an integrated counterexample

On a normalized circle lift the R-071 jet by

$$U_N = z + N^{-1}\sin(Ns)y, \qquad A_{N,t} = a + tN^{-1}\sin(Ns)n. \tag{7.1}$$

Then $DU_N = \cos(Ns)y$, $DA_{N,t} = t\cos(Ns)n$, and the mean local leakage retains the affine slope from (2.9). However

$$\|A_{N,t}\|_{H^2}^2 = 6 + \frac{3t^2}{2}(N^2 + 1 + N^{-2}). \tag{7.2}$$

while the terminal sextic tends to $|z+a|^6 = 27$. The Cameron–Martin cost therefore grows like N^2. This route does not disprove the integrated theorem; it confirms that derivative energy is the necessary missing input.

8 Evidence map and successor

1. **Closed:** exact gauge-kernel classification on every rank stratum, including the doublet and singlet tips.
2. **Closed:** differentiated gauge identity (2.8) and the exact interpretation of the R-071 fixture.
3. **Closed:** inverse-free local bound (3.3) and regularized completion (3.1), with no pseudoinverse at the cone tip.
4. **Closed:** matched same-shell strict-past diagonal leakage with one CM allocation, one terminal sextic allocation, and the single integrable constant (4.18). Its ultraviolet tail is $O(N_{j_0}^{-3})$ and needs only a sixth Gaussian moment.
5. **Retired:** identifying the diagonal with the full terminal leakage. Equations (5.4)–(5.7) expose a nonzero, load-bearing off-diagonal remainder.
6. **Next:** group $\mathcal{O}$ by its largest shell index while retaining the terminal range square and the R-069 control/mixed telescope. Prove a two-parameter Carleson estimate or an exact cancellation compatible with the already-paid R-071 linear model.

9 Devil's-advocate review

1. **DISMISSED: "The R-071 nullvector is an exceptional numerical accident."** Equation (2.3) classifies it as the exact active-doublet gauge direction, and (2.8) reproduces its slope.
2. **VALID with mitigation: "A frame inverse is singular at the tip."** Equation (3.1) inverts only $\vartheta H + \tau I$. The price is the explicit CM allocation $\tau|b|^2/2$.
3. **DISMISSED: "Shellwise Young creates infinitely many constants."** Sequence Holder is applied before one AM–GM step. The only remainder is R_∞, whose expectation is (4.11).
4. **UPHELD against a hidden dimension convention: "The scalar C_D was used for an eighteen-component derivative vector."** Equation (4.4) explicitly defines C_D as the full trace and C_∇ as the full gradient constant.
5. **UPHELD against unit sharp-cube constants: "Y_d is already the terminal sextic."** The actual C_{LP} is retained in (4.14)–(4.18); it is not set to one.

6. **UPHELD against pathwise centering: "Terminal G equals G_j."** Only the expectation identity (4.19) holds, under the same-point value conditioning and real-even regulator.

7. **UPHELD against a missing low boundary: "The high expansion may start at U after deleting lower controls."** Equation (5.1) starts at $z_0 = U + A_{<j_0}$ and retains the R-066 finite-low boundary.

8. **UPHELD against diagonal closure: "The off-diagonal remainder is negligible."** The exact independent fixture has ratio greater than 4087 in (5.7).

9. **UPHELD against full closure: "The diagonal theorem proves controlled-shell one-use."** The triangular off-diagonal range-square estimate, finite-energy extension, umbrella one-use, and Nelson synthesis remain open. Tier stays T4.

Result footer

The primary executable derives the production constant, tests all kernel strata and gauge identities, checks the local bound and regularized completion, verifies the one-constant sequence algebra and Gaussian moment, and executes the exact off-diagonal identity. The non-importing executable rebuilds complex Pauli frames, uses finite-difference derivatives, and checks an independent load-bearing off-diagonal fixture.

```
Result ID: A13-CLASSII-PHASE-KERNEL-CAUSAL-DIAGONAL-ONE-USE-REDUCTION
Precise statement:       The stacked production frame has exactly the local
                         doublet/singlet gauge kernel. The matched strict-past
                         same-shell nonlinear leakage has one-use budgets and
                         an O(N_j0^-3) sixth-moment tail. The terminal leakage
                         equals this diagonal plus an exact nonzero off-
                         diagonal remainder.
Scope:                   Fixed floor; common real-even sharp-cube regulator;
                         regular whole-shell predictable high controls with
                         the finite-low boundary retained.
Dependencies:            A1; R-050; R-066; R-068--R-071.
Evidence grade:          ANALYTIC, EXACT, EXECUTED.
Reproduction command:    python codes/foundations/
                         a13_classii_phase_kernel_causal_diagonal_reduction_verify.py
Expected output:         primary 32/32; independent 26/26; integrated and
                         aggregate counts as pinned in the manifest; exit 0.
Falsification gate:      A kernel stratum, gauge derivative, production
                         constant, one-constant tail, causal boundary,
                         off-diagonal identity, source/PDF hash, or assertion
                         fails.
Tier before / after:     T4 / T4; no promotion.
No-overclaim statement:  The off-diagonal range-square Carleson estimate,
                         finite-energy extension, controlled-shell one-use,
                         Nelson, floor/regulator/volume removal, T5--T7, phase
                         transition, and BCC selection remain open.
Next required action:    Prove the largest-shell triangular two-parameter
                         Carleson estimate for O with the terminal range
                         square and R-069 telescope retained.
```

This result does not construct an interacting measure, remove any regulator, or take infinite volume. It proves no phase transition or BCC selection.